

Часть 1. Поднимаем Postgres

Подготавливаем Dockerfile для нашего постгреса. Подготавливаем compose файл, в котором описываем наш деплой постгреса. Так же добавляем в него Zookeeper, который нужен для непосредственного управления репликацией и определением “лидера” кластера. Создаем postgres0.yml и затем на основе него — postgres1.yml.

собираем и деплоим `docker compose up -d --build`

```
=> [pg-slave internal] load build context      0.0s
=> => transferring context: 3.64kB              0.0s
=> [pg-master 2/5] RUN apt-get update -y && apt-get 600.4s
=> [pg-slave 3/5] RUN python3 -m venv /opt/patroni-venv 12.3s
=> [pg-master 4/5] COPY postgres0.yml /postgres0.yml 0.1s
=> [pg-master 5/5] COPY postgres1.yml /postgres1.yml 0.1s
=> [pg-slave] exporting to image              29.8s
=> => exporting layers                          21.6s
=> => exporting manifest sha256:fb5d32199be2ede67b12b8389 0.0s
=> => exporting config sha256:a7bd1d93c29c451a6fa2398d5d9 0.0s
=> => exporting attestation manifest sha256:f325fc83c58d9 0.1s
=> => exporting manifest list sha256:653fb07a792fb013b18b 0.0s
=> => naming to docker.io/library/patroni:latest 0.0s
=> => unpacking to docker.io/library/patroni:latest 8.0s
=> [pg-master] exporting to image              29.9s
=> => exporting layers                          21.6s
=> => exporting manifest sha256:d48fb25711965ff103f25e8b1 0.0s
=> => exporting config sha256:a22e7ddddd640b10e5b3a7a6b423 0.0s
=> => exporting attestation manifest sha256:42f20e083c3a4 0.1s
=> => exporting manifest list sha256:e25b54475c86bd045210 0.0s
=> => naming to docker.io/library/patroni:latest 0.0s
=> => unpacking to docker.io/library/patroni:latest 8.1s
=> [pg-slave] resolving provenance for metadata file 0.0s
=> [pg-master] resolving provenance for metadata file 0.0s
[+] up 8/8
✓ Image patroni Built 652.8s
✓ Network home_default Created 0.0s
✓ Volume home_pg-master Created 0.0s
✓ Volume home_pg-slave Created 0.0s
✓ Container zoo Created 0.2s
✓ Container pg-master Created 0.2s
✓ Container pg-slave Created 0.2s
✓ Container postgres_entrypoint Created 0.1s
root@pc-network-lab3:/home#
```

Проверяем что зукипер запустился

```
099f8e3124f9 patroni "docker-entrypoint.s..." 31 minutes ago
Restarting (0) 30 seconds ago
pg-master
ce411690791c patroni "docker-entrypoint.s..." 31 minutes ago
Restarting (0) 29 seconds ago
pg-slave
ec58728a7fc9 confluentinc/cp-zookeeper:7.7.1 "/etc/confluent/dock..." 31 minutes ago
Up 31 minutes 2888/tcp, 0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp, 38
88/tcp zoo
○ root@pc-network-lab3:/home#
```

```

root@pc-network-lab3:/home# docker logs zoo
====> User
uid=1000(appuser) gid=1000(appuser) groups=1000(appuser)
====> Configuring
====> Running preflight checks ...
====> Check if /var/lib/zookeeper/data is writable ...
====> Check if /var/lib/zookeeper/log is writable ...
====> Launching ...
====> Launching zookeeper ...
[2025-12-20 13:24:47,480] INFO Reading configuration from: /etc/kafka/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,484] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,484] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,484] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,484] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,487] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-12-20 13:24:47,487] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-12-20 13:24:47,487] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-12-20 13:24:47,487] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.zookeeper.server.quorum.QuorumPeerMain)
[2025-12-20 13:24:47,490] INFO log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2025-12-20 13:24:47,491] INFO Reading configuration from: /etc/kafka/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,491] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,491] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,491] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,491] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-12-20 13:24:47,492] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2025-12-20 13:24:47,500] INFO ServerMetrics initialized with provider org.apache.zookeeper.metrics.impl.DefaultMetricsProvider@1b083826 (org.apache.zookeeper.server.ServerMetrics)
[2025-12-20 13:24:47,513] INFO ACL digest algorithm is: SHA1 (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2025-12-20 13:24:47,513] INFO zookeeper.DigestAuthenticationProvider.enabled = true (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2025-12-20 13:24:47,517] INFO zookeeper.snapshot.trust.empty : false (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,532] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,536] INFO Server environment:zookeeper.version=3.8.4-9316c2a7a97e1666d8f4593f34dd6fc36ecc436c, built on 2024-02-12 22:16 UTC (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,536] INFO Server environment:host.name=zoo (org.apache.zookeeper.server.ZooKeeperServer)
[2025-12-20 13:24:47,798] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2025-12-20 13:24:47,820] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.watch.WatchManagerFactory)
[2025-12-20 13:24:47,820] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.watch.WatchManagerFactory)

```

Проверяем что одна нода из двух стала лидером:

```

pg-slave | 2025-12-20 14:12:20,720 INFO: no action. I am (postgres11), the leader with the lock
pg-master | 2025-12-20 14:12:21,036 INFO: no action. I am (postgres10), a secondary, and following a leader (postgres11)

```

Часть 2. Проверим репликацию

Подключаемся к pg-master и создаем таблицу

```

root@pc-network-lab3:/home# psql -h 127.0.0.1 -p 5434 -U postgres
Password for user postgres:
psql (14.20 (Ubuntu 14.20-0ubuntu0.22.04.1), server 16.11 (Debian 16.11-1.pgdg13+1))
WARNING: psql major version 14, server major version 16.
Some psql features might not work.
Type "help" for help.

postgres=# CREATE TABLE test_table (
    id bigserial PRIMARY KEY,
    name text NOT NULL,
    created_at timestampz DEFAULT now()
);

INSERT INTO test_table(name) VALUES ('hello');
SELECT * FROM test_table;
CREATE TABLE
INSERT 0 1
 id | name |      created_at
-----+-----+-----
  1 | hello | 2025-12-20 14:27:58.623315+00
(1 row)

postgres=#

```

Подключаемся к реплике и видим что данные также появились

```

root@pc-network-lab3:/home# psql -h 127.0.0.1 -p 5433 -U postgres
Password for user postgres:
psql (14.20 (Ubuntu 14.20-0ubuntu0.22.04.1), server 16.11 (Debian 16.11-1.pgdg13+1))
WARNING: psql major version 14, server major version 16.
        Some psql features might not work.
Type "help" for help.

postgres=# SELECT * FROM test_table;
 id | name |          created_at
-----+-----+-----
  1 | hello | 2025-12-20 14:27:58.623315+00
(1 row)

postgres=#

```

Проверяем read-only режим на реплике и он работает

```

postgres=# INSERT INTO test_table(name) VALUES ('hello2');
ERROR:  cannot execute INSERT in a read-only transaction
postgres=#

```

Часть 3. Делаем высокую доступность

Добавляем haproxy в docker-compose.yml, создаем haproxy.cfg и перезапускаем проект и проверяем логи

```

root@pc-network-lab3:/home# docker compose up -d
[+] up 5/5
  Network home_default      Created
  Container zoo              Created
  Container pg-master        Created
  Container pg-slave         Created
  Container postgresql1      Created
root@pc-network-lab3:/home# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c43cf80d8d1    haproxy:3.0                        "/usr/sbin/haproxy -   5 seconds ago Up 3 seconds  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp, 0.0.0.0:7000->7000
6a83b623bc94    patroni                            "docker-entrypoint.s..." 5 seconds ago Up 4 seconds  8008/tcp, 0.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp
71da02cc697b    patroni                            "docker-entrypoint.s..." 5 seconds ago Up 4 seconds  8008/tcp, 0.0.0.0:5434->5432/tcp, [::]:5434->5432/tcp
f6b13d99eece    confluentinc/cp-zookeeper:7.7.1   "/etc/confluent/dock... 5 seconds ago Up 4 seconds  2888/tcp, 0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp, 3888/tcp

```

Логи pg-master

```

2025-12-20 14:39:49.984 UTC [28] LOG:  received promote request
2025-12-20 14:39:49.984 UTC [28] LOG:  redo is not required
server promoting
2025-12-20 14:39:49.989 UTC [28] LOG:  selected new timeline ID: 4
2025-12-20 14:39:50.045 UTC [28] LOG:  archive recovery complete
2025-12-20 14:39:50.054 UTC [26] LOG:  checkpoint starting: force
2025-12-20 14:39:50.058 UTC [24] LOG:  database system is ready to accept connections
2025-12-20 14:39:50.076 UTC [26] LOG:  checkpoint complete: wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed; sync files=2, longest=0.001 s, average=0.001 s; distance=0 kB; estimate=0 kB; lsn=0/6000108, redo lsn=0/6000108
2025-12-20 14:39:50.854 INFO:  establishing a new patroni restapi connection to postgres
2025-12-20 14:39:51.028 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:00.997 INFO:  Lock owner: postgresql0; I am postgresql0
2025-12-20 14:40:01.001 INFO:  Assigning synchronous standby status to ['postgresql1']
2025-12-20 14:40:01.003 UTC [24] LOG:  received SIGHUP, reloading configuration files
2025-12-20 14:40:01.004 UTC [24] LOG:  parameter "synchronous_standby_names" changed to "postgresql"
server signaled
2025-12-20 14:40:01.112 UTC [47] LOG:  standby "postgresql1" is now a synchronous standby with priority 1
2025-12-20 14:40:01.112 UTC [47] STATEMENT:  START_REPLICATION SLOT "postgresql1" 0/6000000 TIMELINE 4
2025-12-20 14:40:03.115 INFO:  Synchronous standby status assigned to ['postgresql1']
2025-12-20 14:40:03.118 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:11.004 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:20.999 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:31.000 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:40.996 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:40:50.996 INFO:  no action. I am (postgresql0), the leader with the lock
2025-12-20 14:41:00.997 INFO:  no action. I am (postgresql0), the leader with the lock
root@pc-network-lab3:/home#

```

Логи zoo

```

[2025-12-20 14:39:48.998] INFO No SessionScavenger set, using defaults (org.eclipse.jetty.server.session)
[2025-12-20 14:39:48.999] INFO node0 Scavenging every 60000ms (org.eclipse.jetty.server.session)
[2025-12-20 14:39:49.002] WARN ServletContext@o.e.j.s.ServletContextHandler@229f66ed{/null,AVAILABLE} has uncovered http methods for path: /* (org.eclipse.jetty.security.SecurityHandler)
[2025-12-20 14:39:49.024] INFO Started o.e.j.s.ServletContextHandler@229f66ed{/null,AVAILABLE} (org.eclipse.jetty.server.handler.ContextHandler)
[2025-12-20 14:39:49.034] INFO Started ServerConnector@564fab8c(HTTP/1.1, (http/1.1){0.0.0.0:8080}) (org.eclipse.jetty.server.AbstractConnector)
[2025-12-20 14:39:49.035] INFO Started @663ms (org.eclipse.jetty.server.Server)
[2025-12-20 14:39:49.035] INFO Started AdminServer on address 0.0.0.0, port 8080 and command URL /commands (org.apache.zookeeper.server.admin.JettyAdminServer)
[2025-12-20 14:39:49.038] INFO Using org.apache.zookeeper.server.NIOServerCnxnFactory as server connection factory (org.apache.zookeeper.server.ServerCnxnFactory)
[2025-12-20 14:39:49.039] WARN maxCnxns is not configured, using default value 0. (org.apache.zookeeper.server.ServerCnxnFactory)
[2025-12-20 14:39:49.040] INFO Configuring NIO connection handler with 10s sessionless connection timeout, 1 selector thread(s), 4 worker threads, and 64 kB direct buffers. (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2025-12-20 14:39:49.041] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2025-12-20 14:39:49.051] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.watch.WatchManagerFactory)
[2025-12-20 14:39:49.055] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.watch.WatchManagerFactory)
[2025-12-20 14:39:49.055] INFO zookeeper.snapshotSizeFactor = 0.33 (org.apache.zookeeper.server.ZKDatabase)

```

Логи haproxy

```
root@pc-network-lab3:/home# docker logs postgres_entrypoint
[NOTICE] (1) : New worker (8) forked
[NOTICE] (1) : Loading success.
[WARNING] (8) : Server postgres/postgresql_pg_master_5432 is DOWN, reason: Layer4 connection problem, info: "Connection refused", check duration: 0ms. 1 active and 0 backup servers left. 0 sessions active, 0 requested, 0 remaining in queue.
[WARNING] (8) : Server postgres/postgresql_pg_slave_5432 is DOWN, reason: Layer4 connection problem, info: "Connection refused", check duration: 0ms. 0 active and 0 backup servers left. 0 sessions active, 0 requested, 0 remaining in queue.
[ALERT] (8) : proxy 'postgres' has no server available!
[WARNING] (8) : Server postgres/postgresql_pg_master_5432 is UP, reason: Layer7 check passed, code: 200, check duration: 1ms. 1 active and 0 backup servers online. 0 sessions requested, 0 total in queue.
root@pc-network-lab3:/home#
```

Подключаемся к новой базе данных и проверяем, что подключение успешное.

Вернулось значение f, значит сервер не находится в режиме восстановления и является мастером.

```
root@pc-network-lab3:/home# psql -h 127.0.0.1 -p 5432 -U postgres
Password for user postgres:
psql (14.20 (Ubuntu 14.20-0ubuntu0.22.04.1), server 16.11 (Debian 16.11-1.pgdg13+1))
WARNING: psql major version 14, server major version 16.
         Some psql features might not work.
Type "help" for help.

postgres=# select pg_is_in_recovery();
 pg_is_in_recovery 
-----
 f
(1 row)

postgres=#
```

Задание:

Отключили доступ до ноды, которая сейчас является мастером и проанализировали логи:

```
root@pc-network-lab3:/home# docker stop pg-master
pg-master
root@pc-network-lab3:/home#
```

По выделенным логам видно, что был получен отказ в соединении с pg-master и что новый мастер - pg-slave готов взаимодействовать.

```
2025-12-20 14:48:31.659 INFO: no action. I am (postgresql1), a secondary, and following a leader (postgresql0)
2025-12-20 14:48:31.877 UTC [34] LOG: replication terminated by primary server
2025-12-20 14:48:31.877 UTC [34] DETAIL: End of WAL reached on timeline 4 at 0/70000A0.
2025-12-20 14:48:31.877 UTC [34] FATAL: could not send end-of-streaming message to primary: server closed the connection unexpectedly
This probably means the server terminated abnormally
before or while processing the request.
no COPY in progress
2025-12-20 14:48:31.884 UTC [29] LOG: invalid record length at 0/70000A0: expected at least 24, got 0
2025-12-20 14:48:31.884 UTC [224] FATAL: could not connect to the primary server: connection to server at "pg-master" (172.18.0.3), port 5432 failed: Connection refused
Is the server running on that host and accepting TCP/IP connections?
2025-12-20 14:48:31.884 UTC [29] LOG: waiting for WAL to become available at 0/7000088
2025-12-20 14:48:32.646 WARNING: Request failed to postgresql0: GET http://pg-master:8008/patroni (HTTPConnectionPool(host='pg-master', port=8008): Max retries exceeded with url: /patroni (Caused by NewConnectionError(HTTPConnection(host='pg-master', port=8008): Failed to establish a new connection: [Errno 111] Connection refused)))
2025-12-20 14:48:32.666 INFO: promoted self to leader by acquiring session lock
server promoting
2025-12-20 14:48:32.668 UTC [29] LOG: received promote request
2025-12-20 14:48:32.668 UTC [29] LOG: redo done at 0/7000028 system usage: CPU: user: 0.00 s, system: 0.00 s, elapsed: 521.96 s
2025-12-20 14:48:32.668 UTC [29] LOG: last completed transaction was at log time 2025-12-20 14:40:01.110696+00
2025-12-20 14:48:32.671 UTC [29] LOG: selected new timeline ID: 5
2025-12-20 14:48:32.747 UTC [29] LOG: archive recovery complete
2025-12-20 14:48:32.757 UTC [27] LOG: checkpoint starting: force
2025-12-20 14:48:32.759 UTC [25] LOG: database system is ready to accept connections
2025-12-20 14:48:32.787 UTC [27] LOG: checkpoint complete: wrote 2 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.006 s, sync=0.002 s, total=0.030 s; sync files=2, longest=0.002 s, average=0.001 s; distance=16383 kB, estimate=16384 kB; lsn=0/7000108, redo lsn=0/7000000
2025-12-20 14:48:33.689 INFO: no action. I am (postgresql1), the leader with the lock
2025-12-20 14:48:43.684 INFO: no action. I am (postgresql1), the leader with the lock
2025-12-20 14:48:53.680 INFO: no action. I am (postgresql1), the leader with the lock
root@pc-network-lab3:/home#
```

Переключили харпоху на нового мастера, проверили, что создание таблицы и запись данных в неё поддерживаются. В итоге после принудительного отключения patroni обеспечил переключение на репликацию.

```

root@pc-network-lab3:/home# psql -h 127.0.0.1 -p 5432 -U postgres
Password for user postgres:
psql (14.20 (Ubuntu 14.20-0ubuntu0.22.04.1), server 16.11 (Debian 16.11-1.pgdg13+1))
WARNING: psql major version 14, server major version 16.
        Some psql features might not work.
Type "help" for help.

postgres=# select pg_is_in_recovery();
 pg_is_in_recovery 
-----
 f
(1 row)

postgres=# █

-----
 f
(1 row)

postgres=# CREATE TABLE test_table (
  id bigserial PRIMARY KEY,
  name text NOT NULL,
  created_at timestamptz DEFAULT now()
);

INSERT INTO test_table(name) VALUES ('hello');
SELECT * FROM test_table;
ERROR:  relation "test_table" already exists
INSERT 0 1
  id | name |          created_at
-----+-----+-----
  1 | hello | 2025-12-20 14:27:58.623315+00
  2 | hello | 2025-12-20 14:52:02.166307+00
(2 rows)

postgres=# █

```

Ответы на вопросы:

1. Порты 8008 и 5432 вынесены в разные директивы, *expose* и *ports*. По сути, если записать 8008 в *ports*, то он тоже станет *exposed*. В чем разница?

Expose: просто указывает, что контейнер использует порт, но не делает его доступным для внешнего мира.

Ports: пробрасывает порт контейнера на хост, делая его доступным для внешних expose.

Expose — это только документация, а ports — это настройка доступности порта для внешнего мира.

2. При обычном перезапуске композ-проекта, будет ли сбилден заново образ? А если предварительно отредактировать файлы *postgresX.yml*? А если содержимое самого *Dockerfile*? Почему?

При выполнении команды *docker-compose up* используются уже существующие образы. Пересборка образов происходит только при указании флага *--build* или если необходимого образа нет в локальном хранилище.

Docker проверяет наличие образов, указанных в *docker-compose.yml*, и, если они существуют, запускает контейнеры на их основе.

При изменении файлов конфигурации образы не пересобираются. Файлы конфигураций описывают конфигурацию запуска контейнеров, но не содержат инструкций для сборки образов. Если образы уже существуют, Docker

использует их без изменений. Перезапуск проекта с помощью `docker-compose up` просто переопределяет параметры запуска контейнеров на основе обновленной конфигурации.

Если внесены изменения в `Dockerfile`, образы пересобираются только при явном указании: флага `--build` при запуске или при ручной пересборке образа командой `docker-compose up build`.

`Dockerfile` содержит инструкции для создания образов, но Docker не отслеживает изменения в файле автоматически. Чтобы применить изменения, необходимо вручную запустить процесс пересборки образа.